

Kareem Hesham

Python Automation, Web Scraping & Backend Engineer

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PROFILE

Python engineer building production-grade automation systems, backend APIs, and data acquisition pipelines. Specializes in request-level automation and protocol analysis — reading APIs directly instead of driving a browser, which is what made a peak day of one million records from ten thousand requests possible. Published security researcher; three independently verified professional references.

EXPERIENCE

Independent Python Automation & Backend Engineer 2022 – PRESENT

Freelance · Contract · Cairo, Egypt · Remote · Automation · Data acquisition · Reverse engineering · Backend APIs

Contract engagements rather than salaried roles — embedded in teams of roughly two to six engineers alongside data scientists and ML engineers, delivering against their requirements and review.

- ▶ Build request-level automation that reproduces protected client behavior without a browser or DOM — one engagement delivered 1M records in a day from ~10K direct HTTP requests.
- ▶ Reverse engineer client-side signatures, dynamic tokens, fingerprint challenges, and Android application flows across 600+ distinct targets.
- ▶ Handle bot-mitigation and verification flows across roughly 30 challenge types, working only on client-authorized targets and strictly within the contracted engagement scope.
- ▶ Diagnose and ship fixes within 6–24 hours when a target changes its protections, keeping delivery pipelines running.
- ▶ Design typed FastAPI services with layered architecture, PostgreSQL persistence, Alembic migrations, JWT authentication, and pytest coverage.
- ▶ Develop resilient acquisition pipelines with session control, proxy rotation, and rate-limit handling — largest single delivery was 60GB across 35M validated rows.
- ▶ Collaborate with data scientists and ML engineers on scoped deliveries, coordinating tasks and shipping on schedule.
- ▶ Turn investigation tooling into documented open-source utilities and publish the underlying technical research.

All work was performed under client contract, on systems the client was authorized to access and within the agreed scope. Figures span engagements between 2022 and 2026; client names and target systems are covered by confidentiality agreements, with specifics available on request under NDA.

SELECTED OPEN SOURCE

capture2tls Python /capture2tls
A zero-dependency converter from raw proxy captures to fingerprint-aware Python request code.

jsunveil JavaScript /jsunveil
A pattern-based JavaScript deobfuscator that survives regenerated identifiers, alphabets, and rotation offsets.

API Automation Platform Python REFERENCE PROJECT /api-automation-platform
A production-minded FastAPI foundation for authenticated API integrations and background synchronization workflows.

SECURITY RESEARCH

I Spent a Week Inside a Top-Tier Web Anti-Bot Stack
A practical teardown of layered anti-bot infrastructure: rotating JavaScript, device fingerprinting, behavioral signals, and reproducing a headless challenge outside the browser.

An Infinite Loop in Samsung's UltraHDR Stack
A CWE-835 denial-of-service finding traced through arm64 reverse engineering and validated through coverage-guided fuzzing with AFL++.

TECHNICAL STACK

- Backend**
Python, FastAPI, SQLAlchemy, PostgreSQL, Alembic, REST APIs, JWT, Pydantic
- Automation**
Playwright, Selenium, Scrapy, requests, asyncio, CDP
- Networking**
HTTP, HTTP/2, TLS, Cookies, Sessions, Headers, Proxy Rotation, Burp Suite, Charles
- Reverse Engineering**
JavaScript, Node.js, Babel, AST, Frida, JADX, Android
- Infrastructure**
Docker, Linux, GitHub Actions, pytest, Git, CI/CD

REFERENCES

Three professional references — a data scientist, an ML engineer, and a web scraper who worked directly on these deliveries — independently collected and identity-verified.

Verified profile available at app.zincwork.com — full link on the portfolio site.

EDUCATION

Bachelor of Laws (LL.B.)
Cairo University · 2018 — 2023

Lets me read a terms of service, DPA, or engagement scope and locate the line before code is written. The underlying habit — close reading under adversarial conditions — transfers to protocol analysis.

CERTIFICATIONS

- Build REST APIs with FastAPI
LinkedIn Learning
- Practical GitHub Actions
GitHub and LinkedIn Learning
- Docker Foundations Professional Certificate
Docker and LinkedIn Learning

LANGUAGES

- Arabic — Native
- English — Professional working proficiency